Race, explain and transfer

antocciphysics | TEACHER KEY | CPA

Learning target: Evaluate a model against measured performance and defend design decisions.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 8, Day 3; 6 course days later.*
Ideal speed becomes 6 times as large. Actual rpm, load or slipping may change.

02. Explain a worked example (worked)
Electrical input to motor/mechanical motion plus thermal/sound transfers; actual measurements distinguished from illustrative values.

03. Look and explain (model)
Three supported explanations using relevant mechanics/circuit ideas.

04. Build the solution (practice)
.50 m/s; 16.7% below the prediction.

05. Use the idea (reason)
Actual evidence and bounded explanations; a lower race placing alone does not diagnose a cause.

06. Choose a method (mixed) *Review: Unit 7, Day 15; 12 course days later.*
A generator. Some energy can become heating, friction or sound.

07. Exit check (exit)
Student names a concrete next step tied to evidence or a skill; self-rating is not a teacher score.

08. STRETCH: Apply it another way (stretch)
Example lower gearing may improve starting torque but reduce unloaded speed; test start reliability and timed speed under controlled conditions.

Teaching note: Keep the official race/judging record separate from these supplied practice examples. Report your own results honestly.

Race, explain and transfer

antocciphysics | TEACHER KEY | HONORS

Learning target: Evaluate a model against measured performance and defend design decisions.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 8, Day 3; 6 course days later.*
Torque ratio 8; speed ratio 1/8. Assumes ideal gears with negligible losses.

02. Explain a worked example (worked)
Electrical input to motor/mechanical motion plus thermal/sound transfers; actual measurements distinguished from illustrative values.

03. Represent and interpret (model)
Three supported explanations using relevant mechanics/circuit ideas.

04. Solve and justify (practice)
.75 m/s; 16.7% below prediction; load, slip, friction or model assumptions, with evidence needed to attribute cause.

05. Analyze the reasoning (reason)
Actual evidence and bounded explanations; a lower race placing alone does not diagnose a cause.

06. Choose a method (mixed) *Review: Unit 7, Day 15; 12 course days later.*
Combined fraction=0.4608 or 46.08%, multiplying the two efficiencies.

07. Exit check (exit)
Ideal speed doubles; increased required torque, loading, slip and clearance may change actual rpm/performance.

08. STRETCH: Apply it another way (stretch)
Input 24 J; K 1 J; ratio≈4.17%. Also account for rotational energy, work against resistance, stored energy and time history.

Teaching note: Keep the official race/judging record separate from these supplied practice examples. Report your own results honestly.